

GRAPHING PROPORTIONAL RELATIONSHIPS



LESSON INTRODUCTION

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

LEARNING TARGETS

- I can graph a proportional relationship from a table, an equation, or a written description.
- I can describe common features of the graphs of proportional relationships

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

ESSENTIAL QUESTIONS

- What is a proportional relationship?
- What are common features of the graphs of proportional relationships?

LESSON NARRATIVE

The lesson formally introduces proportional relationships with a focus on their graphical representations. Students will use graphs and tables to explore proportional relationships, their graphs, and their features in real-world contexts. The 11.1.2 Exploration is designed to activate students' prior knowledge about unit rate, proportional relationships, and points on the coordinate plane. After completing the 11.1.2 Exploration, students summarize what they discovered by completing statements to describe the key features of proportional relationships.

COMPONENT SUMMARY

11.1.1 WARM-UP (~5-10 MIN.)

Consider connections between math and language

11.1.2 EXPLORATION (20-25 MIN.)

Explore proportional relationships and their graphs

11.1.3 BRING IT TOGETHER (5-10 MIN.)

Describe common features of the graphs and proportional relationships

11.1.4 WRAP-UP (~5 MIN.)

Graph a proportional relationship after completing a table of values

RESOURCES

GLOSSARY

Key Terms:

- Proportional Relationship

Additional Terms: N/A

MATERIALS

- (Optional) Graph paper
- Dancing with Math video

ADDITIONAL PREPARATION

- 11.1.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may graph proportional relationships incorrectly, confusing the x and y values.
- Students may not understand what the unit rate represents in context.

THINGS TO CONSIDER

- Consider encouraging students to label the columns of the tables with x and y to ensure they graph each point correctly.
- It may be helpful to have students verbalize what the unit rate represents in each scenario to ensure they understand what the unit rate represents.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

11.1.2 Exploration provides an opportunity for students to engage in a Notice and Wonder routine. Encourage students to notice mathematically significant features of the graphs to make connections between graphs and solidify understanding of the features of proportional relationships on graphs. This will provide an entry point for all students in a low-stakes atmosphere.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Throughout this lesson, proportional relationships are presented in the context of real-world scenarios. This provides an opportunity for students to connect mathematical concepts to everyday experiences, increasing student investment and engagement with the content.

DIFFERENTIATE INSTRUCTION

This lesson provides a variety of visual entry points for students to engage with the content. Specifically, students utilize graphs and tables to display proportional relationships. To provide additional visual entry points as students work to determine the corresponding values in the tables, encourage students to use arrows to illustrate the direction of multiplication and label the unit rate to ensure they are multiplying by the correct value.

n length (in.)		c length (cm)
1	$\times 2.54$ →	2.54
6	$\times 2.54$ →	15.24
$3\frac{1}{2}$	$\times 2.54$ →	8.89
2.25	$\times 2.54$ →	5.715

11.1.1 WARM-UP: DANCING WITH MATH

QUESTIONING

This video provides an opportunity to engage students through their own experiences of speaking different languages or living in another country. Consider having students discuss their own experiences to connect to Karin's to increase investment and engagement.



11.1.1 Warm-Up: Dancing with Math

Karin Ditchfield lived in three different countries growing up – Brazil, England, and Argentina – where three different languages are spoken. Learning new languages can be challenging, but she explains despite these challenges the language of math was always the same.

1. Karin says, "Math is not just about working with numbers. I see it as a global language that has a positive impact on the world."

What are some reasons why math could be considered a global language?

Answers will vary. Even if you have different words to describe something, 1 is still 1, $2 + 2$ will always equal 4, and 12^2 will always give you 144 no matter where you live or what language you speak.

2. Karin mentioned that growing up in Brazil meant growing up dancing. She says, "When you're learning a new dance, practice makes perfect. It's exactly the same with math."

Explain whether you agree with this explanation.

Answers will vary. When you're learning a dance, you will make many mistakes before getting the steps right. In math, you will make mistakes before mastering a new concept.



11.1.2 Exploration: Graphs of Proportional Relationships

In this course, **proportional relationships** have been used to make predictions about populations from sample data, create scale drawings, convert between units of measure, and solve problems involving percentages.



A **proportional relationship** is a collection of pairs of numbers that are in equivalent ratios.

Proportional relationships can be represented by graphs, tables of values, equations, and written descriptions.

Work with your partner to complete the following.

1. Madelyn and her family went to a U-Pick farm in Lithia, FL to pick blueberries. The table shows the weight, in pounds (lb), and price, in dollars, of the bag of blueberries each family member picked.

Weight (lb)	Price (\$)
0.75	2.85
2.6	9.88
2	7.60
1.2	4.56

- a. What is the unit price per pound of blueberries?
The unit price is \$3.80 per pound.
- b. Explain how you determined the unit price.
To determine the unit price, I divided the dollar amounts by the pounds to find the price per pound.

11.1.2 EXPLORATION: GRAPHS OF PROPORTIONAL RELATIONSHIPS

TEACHER MOVES

11.1.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the Exploration as they engage with each new situation.

Observe and listen as students work with their partner. Make note of student ideas that you can ask them to share in the whole group in 11.1.3 Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Exploration.

TEACHER MOVES

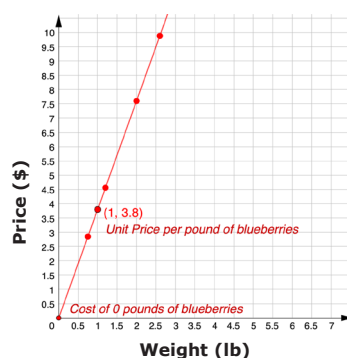
Question 1 is intended to spiral from prior units, as students have worked with unit rates before. This section is also intended to begin to lay the foundation for finding the constant of proportionality in future lessons in this unit without explicitly naming it.

11.1.2 EXPLORATION: GRAPHS OF PROPORTIONAL RELATIONSHIPS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Graphing the data from the tables provides a visual representation of the proportional relationships presented in this Exploration.

- c. Graph the relationship by plotting a point to represent each family member's bag of blueberries.



- d. Plot a point on the graph to represent the unit price per pound of blueberries.
Labeled on graph
- e. Plot a point on the graph to represent the price for 0 pounds of blueberries.
Labeled on graph
- f. Connect the points plotted on the graph. What do you notice?
The points form a straight line, beginning at the origin. Other points are also included on the line.
- g. Discuss with your partner whether the graph of the relationship between the number of pounds of blueberries picked and price extends beyond the points plotted on the graph in either direction.

DIFFERENTIATE INSTRUCTION

To provide an additional visual entry point, encourage students to draw an arrow from the first column to the second and label the multiplication ($\times 2.54$). This will help students keep track of each row in the table as they fill in the length column.

- h. Complete the statements.

The graph ☐ does ☒ does not extend below the point (0, 0) because it is not possible to pick a ☐ positive ☒ negative number of pounds of blueberries.

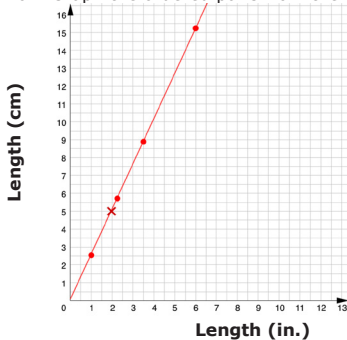
The graph ☐ does ☒ does not extend above the point (2.6, 9.88) because it is possible to pick any ☐ positive ☒ negative number of pounds of blueberries.

2. There are approximately 2.54 centimeters (cm) in every 1 inch (in.). This proportional relationship can be represented by the equation $c = 2.54n$, where c represents the length of an object in centimeters, and n represents the length of the same object in inches.

- a. Use the equation to complete the table of values.

n length (in.)	c length (cm)
1	2.54
6	15.24
$3\frac{1}{2}$	8.89
2.25	5.715

- b. Graph the ordered pairs from the table of values.



- c. Connect the points plotted on the graph to create a ray that represents all possible equivalent lengths.
- d. Discuss with your partner whether there is a point on the line that represents the approximate number of inches that are equivalent to 5 centimeters. If there is such a point, draw an X to mark this point on the line. If there is not such a point, explain why not.
Labeled on graph
3. An empty bathtub is filling at a rate of 4.5 gallons per minute.
- How many gallons of water are in the tub after 0 minutes?
There are 0 gallons of water after 0 minutes.
 - How many gallons of water are in the tub after 3 minutes?
There are 13.5 gallons of water after 3 minutes.

11.1.2 EXPLORATION: GRAPHS OF PROPORTIONAL RELATIONSHIPS (CONTINUED)

TEACHER MOVES

Notice and Wonder

Engage students in a Notice and Wonder routine after they have completed and labeled both graphs. Ask guiding questions to encourage students to draw conclusions about the features of proportional relationships when graphed:

- What do you notice about the graph of the relationship between the price and weight of blueberries? Between the lengths in inches versus centimeters? (The line is straight; the line goes through the origin.)
- What does this make you wonder about graphing proportional relationships? (Will the line always be straight? Can proportional relationships have negative values? Does it have to go through the origin?)

ENRICHMENT

Students may struggle to determine the number of gallons of water in the tub and vice versa. Ask questions to further scaffold if necessary.

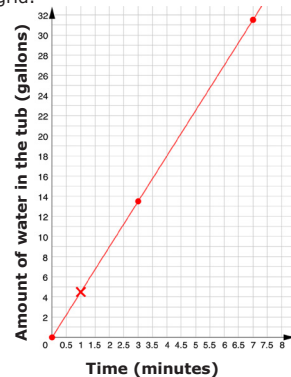
- What is the unit rate in gallons of water per minute? (4.5 gallons per minute.)
- How can you use the unit rate to determine the number of gallons of water in the tub after 3 minutes? (I can multiply the unit rate, 4.5, times the number of minutes, 3.)
- How can you determine the number of minutes it takes to fill the tub with 31.5 gallons of water? (Since I multiplied to find the number of gallons, I can perform the opposite operation to find the number of minutes. I can divide by 4.5.)

11.1.2 EXPLORATION: GRAPHS OF PROPORTIONAL RELATIONSHIPS (CONTINUED)

- c. How many minutes does it take for the tub to fill to 31.5 gallons of water?

It takes 7 minutes for the tub to fill up with 31.5 gallons.

- d. Using the values determined in Parts A-C, represent the relationship of the amount of water in the tub, in gallons, with time, in minutes, on the coordinate grid.



- e. Connect the points plotted on the graph to create a ray.

- f. Complete the statement.
The bathtub is filling up at a constant unit rate of

4.5

☐ gallons
☐ minutes

per

☐ gallon.
☐ minute.

- g. Place an X on the graph at the point that represents the unit rate described in Part F.

Labeled on graph



11.1.3 Bring It Together: Graphs of Proportional Relationships

Each graph created in the Exploration represents a proportional relationship. Graphs representing proportional relationships have certain features in common.

1. Complete each statement to describe the common features of the graphs of proportional relationships.
 - a. Graphs of proportional relationships

☒ include
☐ do not include

 the origin, (0,0), because when the value of one quantity equals 0, the value of the other quantity

☒ will also
☐ will not

 equal 0.
 - b. Graphs of proportional relationships form
 - a

☒ line.
☐ curve.
 - c. A unit rate for the proportional relationship can be identified on the graph using the

☐ x-value
☒ y-value

 of the point where the

☒ x-value
☐ y-value

 is 1.

11.1.3 BRING IT TOGETHER: GRAPHS OF PROPORTIONAL RELATIONSHIPS

TEACHER MOVES

Consider allowing students to discuss each statement with their partner first before facilitating a whole group debrief to ensure students are correctly summarizing the key features of the graphs of proportional relationships.

TEACHER MOVES

There are two correct answers for question 1c, depending on what students select for the first choice. This is intentional and will be further explored in the next lesson.

QUESTIONING

For students who finish early, consider having them use graph paper to create a graph that illustrates a proportional relationship. This could include an example and nonexamples.



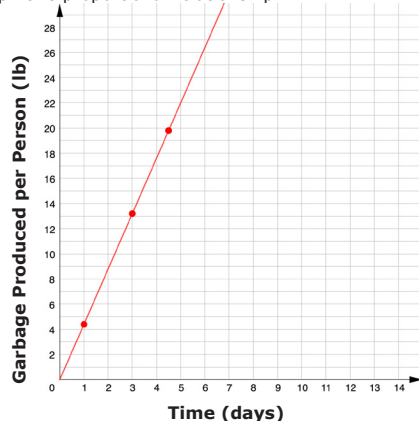
11.1.4 Wrap-Up: Ticket Out the Door

1. The Environmental Protection Agency (EPA) estimated that in 2013, the average amount of garbage produced in the U.S. was 4.4 pounds (lb) per person per day.

The table of values represents the proportional relationship between the amount of garbage produced in the U.S., in pounds, per person, with time, in days.

Number of Days	Garbage per Person (lb)
1	4.4
3	13.2
4.5	19.8
5	22

Graph this proportional relationship.



11.1.4 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

The information provided from the students here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.